ثانوية المدينة التعليمية
Education City High School

1.14

Function Model Construction and Application

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ______________________ Class: ____________ Date: ____________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Choose a model family before solving for parameters.

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2. Write one precise sentence explaining this principle: Each independent condition usually supplies one equation for one unknown parameter.

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3. Write one precise sentence explaining this principle: Parameters should be interpreted with units and contextual meaning.

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4. Write one precise sentence explaining this principle: Model output is an estimate whose reliability depends on assumptions and distance from observed data.

Core

5. Construct a linear model through $(2, 11)$ and $(7, 31)$.

6. A quadratic has vertex $(3, -2)$ and passes through $(5, 6)$. Find it.

7. An exponential model has $f(0) = 12$ and $f(4) = 30$. Find the four-unit growth factor.

8. State what the formula linear: $f(x) = mx + b$ tells you and name any required restriction.

9. State what the formula quadratic: $f(x) = a(x-h)^2+k$ tells you and name any required restriction.

10. State what the formula exponential: $f(x) = ab^x$ tells you and name any required restriction.

Medium

11. Construct the model: linear through (1,7),(5,23).

12. Construct the model: quadratic vertex (2,-1), through (4,11).

13. Construct the model: exponential with f(0)=8,f(2)=18.

14. Construct the model: direct variation with y=35 when x=5.

15. Construct the model: inverse variation with y=4 when x=6.

Hard

16. A quadratic has zeros -2 and 5 and passes through (0,-20). Find it.

17. An exponential model has $A(3) = 40$ and doubles every 6 units. Write it using time origin 3.

18. A taxi fare is 12 QAR plus 2.5 QAR/km, but trips are capped at 80 QAR. Write a model.

19. Why may changing the time origin improve parameter interpretation without changing predictions?

Difficult

20. Find a cubic $P(x) = ax^3 + bx^2 + cx + d$ with zeros 0 and 2, where 2 is double, and P(1)=3.

21. Find an exponential model through $(2, 12)$ and $(7, 30)$, then solve when it reaches 50.

22. A model $f(x) = a/(x - h) + k$ has asymptotes x=2,y=5 and passes through (3,1). Find a.

Challenge / HOT

23. Construct two different model families that agree at x=0 and x=1 but diverge afterward.

24. Explain why n independent parameters generally require n independent conditions, and give a counterexample when conditions are dependent.

AP-Style Multiple Choice

25. Line through (0,3),(2,11) is

- **(A)** $4x + 3$
- **(B)** $3x + 4$
- **(C)** $8x + 3$

(D) $4x - 3$

26. Vertex form is best when given
- **(A)** zeros
- **(B)** vertex and a point
- **(C)** end behavior only
- **(D)** a table with ratios

27. An exponential through (0,a) has a equal to
- **(A)** base
- **(B)** initial value
- **(C)** growth rate only
- **(D)** asymptote

28. Interpolation means prediction
- **(A)** inside observed input range
- **(B)** far beyond data
- **(C)** only at x=0
- **(D)** without a model

29. Which parameter has units output/input in $mx + b$?
- **(A)** m
- **(B)** b
- **(C)** x
- **(D)** both m and b

30. If a model has more unknown parameters than independent conditions, it is usually
- **(A)** overdetermined
- **(B)** underdetermined
- **(C)** unique
- **(D)** impossible always

AP-Style Free Response

31. A ball height follows a quadratic with zeros at t=0 and t=8 and maximum height 32. (a) Construct the model. (b) Find height at t=3. (c) Interpret domain.

32. A quantity is 120 at t=0 and 75 at t=10. Assume exponential decay. (a) Construct model. (b) Find half-life. (c) Predict t when amount is 20. (d) State a limitation.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.